Ziqi CHEN

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Research Interests: Efficient & Adaptive LLMs, Multi-Agent Collaboration, Reinforcement Learning

EDUCATION

The Chinese University of Hong Kong, Shenzhen (CUHK-Shenzhen) Shenzhen, China
B.S. in Computer Science and Engineering Sep. 2023 – May 2027 (Expected)
CGPA: 3.69/4.00 | GRE: 329 + 3.5 | TOEFL: 111

PUBLICATIONS & PREPRINTS

SciTestBench: Can AI Agents Find Errors in Scientific Software?
Ziqi Chen, Tian Li, Kexin Pei.
Under review, 2026. [Website]

The Catfish Effect: Heterogeneous Competition in Group-Relative Learning
Yiyang Lu, **Ziqi Chen**, Xilin Xia, Hongyuan Zha.
Under review, 2026.

DocMaster: A Hierarchical Structure-Aware System for Document Analysis
Ziqi Chen, Yingli Zhou, Fangyuan Zhang, Quanqing Xu, Chuanhui Yang, Yixiang Fang.
arXiv preprint, 2026. [Website]

RESEARCH EXPERIENCE

Research Intern, University of Chicago Apr. 2026 - Present
Advisors: Prof. Kexin Pei and Prof. Tian Li

- **SciTestBench: Can AI Agents Find Errors in Scientific Software?**
  - Led the development of a benchmark evaluating whether coding agents can detect errors in scientific software, and proposed *scientific sanitizers*, runtime checks that verify whether program behaviors satisfy domain-specific scientific principles.
  - Constructed **149** issue-reproduction tasks and **400** scientific sanitizers across **6** scientific domains; evaluated **8** frontier coding agents, with the best (Claude Code + Opus 5.5) achieving only a **56.4%** success rate.
  - Identified and reported **56** previously unknown bugs in open-source scientific software, with **31** confirmed and **9** fixed by maintainers.

Research Intern, CUHK-Shenzhen Mar. 2026 – Sep. 2026
Advisor: Prof. Hongyuan Zha

- **The Catfish Effect: Heterogeneous Competition in Group-Relative Learning**
  - Proposed Catfish GRPO, a group-relative RL framework that introduces a heterogeneous competitor via scalar outcomes only, to investigate how heterogeneous competition reshapes policy training.
  - Designed a *virtual catfish* that replaces real LLM competitors with a binary outcome model, improving performance without additional LLM inference cost.
  - Evaluated on Qwen and Llama models (3B–8B) across math and multi-hop QA tasks, improving accuracy over vanilla GRPO by **+2.07** pp on GSM8K and **+3.05** pp on MATH-500.

Research Intern, CUHK-Shenzhen Feb. 2025 – Mar. 2026

Advisors: Prof. Benyou Wang and Dr. Yan Hu

- **Adaptive Routing for Multi-Timescale Memory in Language Models**
  - Proposed a token-level routing mechanism that dynamically selects read/write depths in multi-layer memory modules, improving memory update efficiency and mitigating forgetting.
  - Designed preliminary experiments on a four-layer memory module, revealing that tokens contribute unevenly across memory layers, and motivating adaptive computation for memory management.
- **GRouter: A Learned Model Routing System for Multi-Agent Systems**
  - Proposed a learned router that assigns a suitable base LLM to each agent in a multi-agent system to optimize the cost–accuracy trade-off for different tasks.
  - Collected **8,000** multi-agent trajectories on GSM8K and HumanEval to train a lightweight MLP router, reducing token cost by **4.2%** while preserving **90.3%** accuracy.
- **G-Designer-Pro: Optimizing Multi-Agent Communication Topologies via GRPO**
  - Enhanced G-Designer, a multi-agent communication-topology generation method, by replacing PPO with GRPO and introducing a utility-based reward to stabilize training.
  - Reduced token cost by **9.7%** while maintaining **89.1%** accuracy on GSM8K and HumanEval.

INDUSTRY EXPERIENCE

Founding Engineer, Mortar Dec. 2025 – Feb. 2026

Led the software engineering team in building a data-driven AI platform for company operations monitoring and risk analysis using Vercel and Supabase. Secured **$30K** in startup programs from Anthropic and NVIDIA.

NLP Engineer Intern, Lalamove May 2025 – Aug. 2025

Fine-tuned a multilingual BERT model on customer-service QA dialogues, achieving **95.7%** consistency with the existing third-party service it replaced while reducing annual costs by **$50K**.

PROFESSIONAL SERVICE

Reviewer, ICLR 2026 Workshop on AI with Recursive Self-Improvement Mar. 2026

Presenter, The 1st CUHK-Shenzhen Undergraduate Research Conference Mar. 2026

TEACHING

- AIE2040 Statistical Foundations for AI Fall 2026
- CSC1004 Computational Laboratory Using Java Spring 2026
- CSC1001 Introduction to Computer Science and Java Programming Fall 2025

AWARDS & HONORS

- UChicago Student Summer Research Fellowship ($5,000) Apr. 2026
- Undergraduate Research Award, CUHK-Shenzhen Mar. 2026
- ShanHaiWoo Scholarship ($1,000) Aug. 2025
- MCM/ICM, Honorable Mention May 2025
- GBA International Programming Contest, First Prize Apr. 2025
- Dean's List 2023 – 2026

OTHER

- **Communities:** CAMEL Contributor, Sentient Ambassador, Obsidian Plugin Developer